

3. The photograph shows a single leaf of white clover (*Trifolium repens*). Each leaf has three leaflets.



Some white clover plants can produce cyanide. Those plants that can produce cyanide are called cyanogenic; those that cannot produce cyanide are called acyanogenic.

The leaves of cyanogenic plants can be damaged by slugs or by ice crystals when exposed to low temperatures. When cell membranes are damaged cyanide is released. Cyanide is toxic to the cells of animals.

(a) State why it is an adaptation for the clover to produce cyanide.

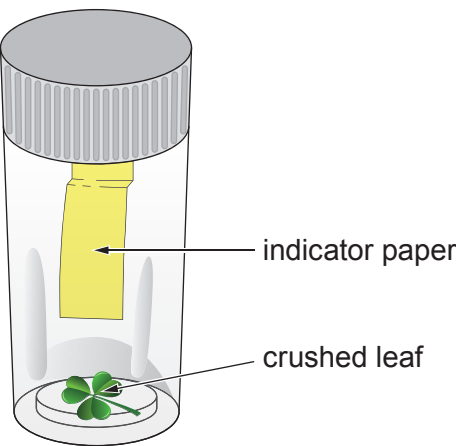
[1]

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- (b) To test different white clover plants for cyanide production, a single leaf from each plant is crushed in a few drops of water at the bottom of separate 50 mm × 15 mm glass tubes. A short slip of indicator paper is suspended in each tube. Each tube is then incubated for 2 hours at 40 °C. If cyanide is released it will turn the indicator paper from yellow to a reddish brown.



- (i) Identify **two** controlled variables in this test. [2]

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- (ii) Explain why the leaves are crushed before inserting the indicator paper. [1]

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- (iii) The indicator paper contains a corrosive chemical. Describe a risk in performing this test and the control measure that should be taken to minimise the risk. [1]

Risk

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Control Measure

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- (c) A researcher performed an investigation to discover if there was a correlation between altitude (distance above sea level) and the percentage of cyanogenic clover in a population.

A large number of sites were sampled at altitudes between 20 m and 240 m above sea level. Each site covered an area of 10 m × 10 m. At each site clover leaves were collected at random. Each leaf was then tested for cyanide production.

- (i) Identify the independent and dependent variables in this investigation. [1]

Independent variable:

Dependent variable:

- (ii) Describe a method that could be used at each site to collect clover leaves at random. [3]

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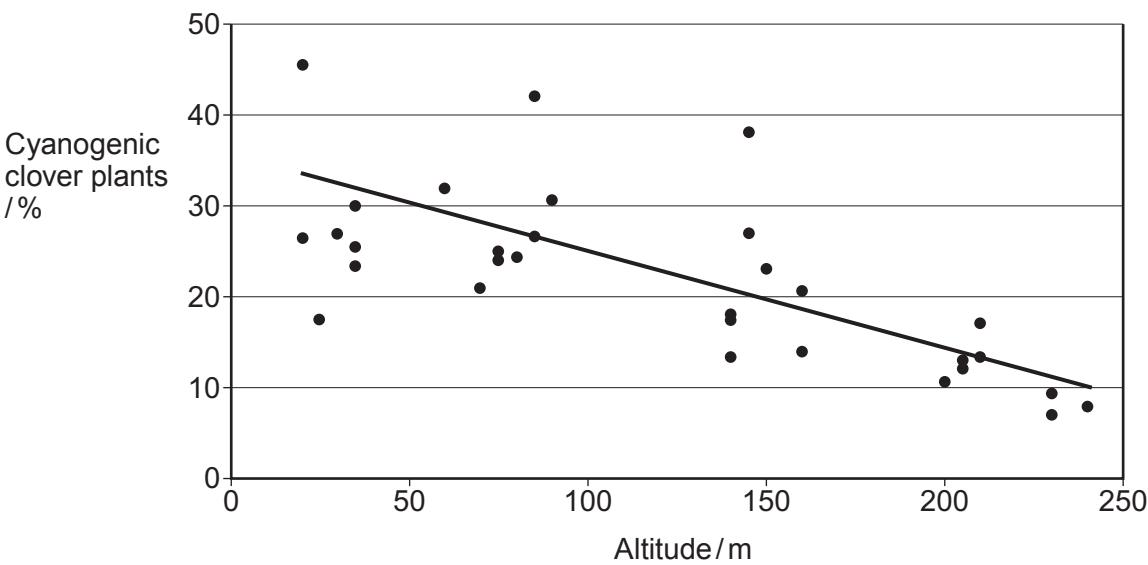
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(d) The scatter diagram summarises the results of this investigation.



The researcher claimed that the reason for the change in the percentage of cyanogenic clover plants was due to differences in **environmental temperature**. Was the researcher justified in making this claim? Give a reason for your answer. [1]

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(e) Cyanogenesis in *Trifolium repens* is an example of genetic polymorphism. Explain what is meant by genetic polymorphism. [2]

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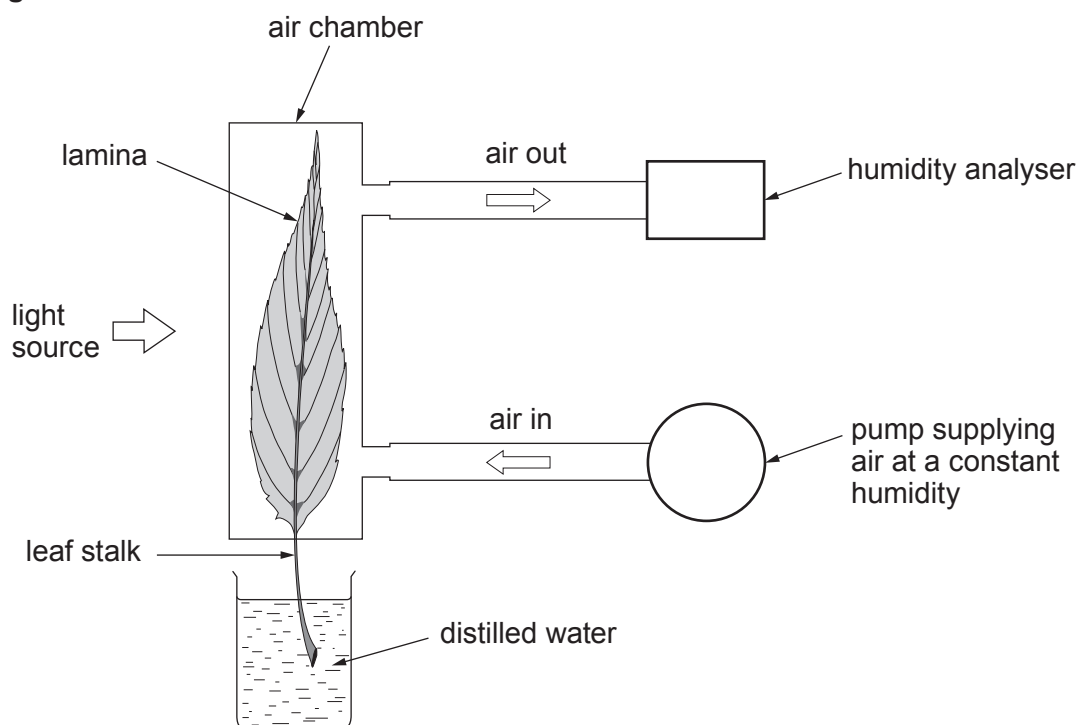
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- (c) Abscissic acid (ABA) is a plant hormone that is produced in roots in response to soil water shortage. It is then transported to guard cells. An investigation was carried out to determine the effect of ABA on the humidity of air surrounding a bean leaf (*Phaseolus vulgaris*).

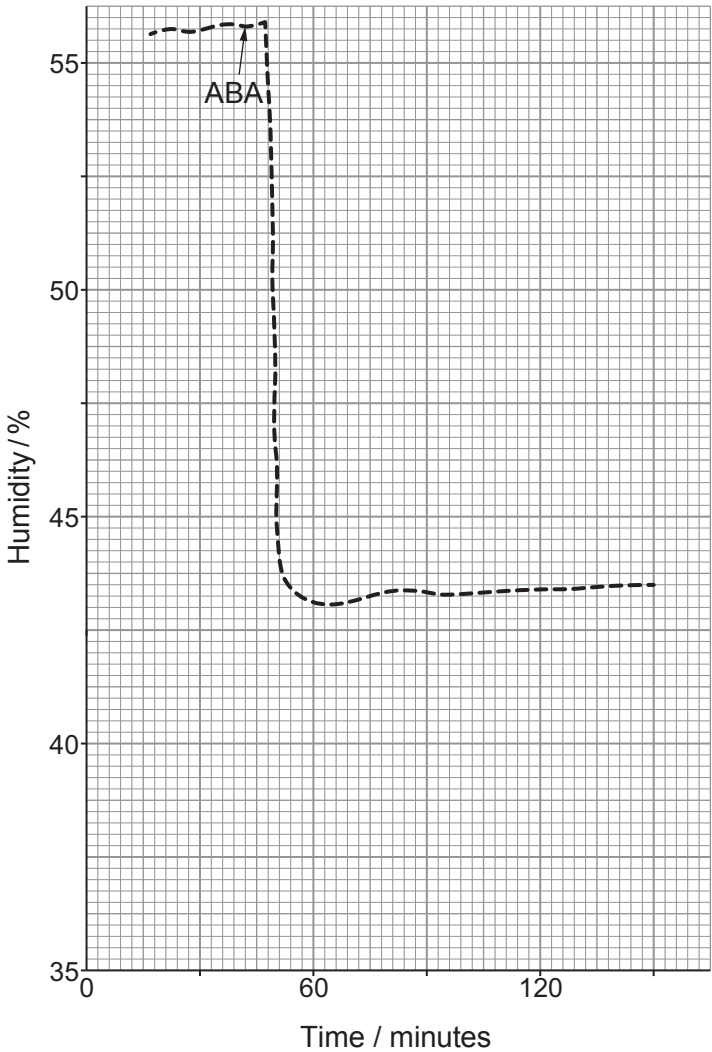
A bean leaf stalk was cut underwater and the leaf inserted into an illuminated, transparent chamber. The lamina of the experimental leaf was held at right angles to the light source. The leaf was supplied with air at a constant humidity. The humidity of the air leaving the chamber was then measured using an analyser. The apparatus used is shown in **Diagram 3**.

Diagram 3



After 40 minutes, ABA was added to the distilled water. The results are shown in **Graph 1**.

Graph 1



(i) Explain why the bean leaf stalk was cut underwater. [2]

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(ii) Describe and explain the change in the humidity of the air leaving the leaf chamber **after** the addition of ABA. Explain how this change benefits the bean plant. [4]

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